Meeting 3 of 7 · October 13, 2026

IMPLEMENTATION PROGRESS

From data to a working quantum classifier

Data pipeline · classical baselines · frozen VQC · encoding repair · first noise characterization

Today: implementation milestones completed Oct 6–7, 2026 (Phases 1–3B). The whole study, including the one-time sealed test, was completed on Oct 8; later meetings present those results in depth.

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Proposed split: Mayank (slides 1–3) · Tushar (slides 4–5) · about 5 minutes



Leakage-safe data and strong classical baselines

284,807 transactions
492 frauds · public ULB data

→

Exact deduplication
1,081 duplicate rows removed (773 groups)

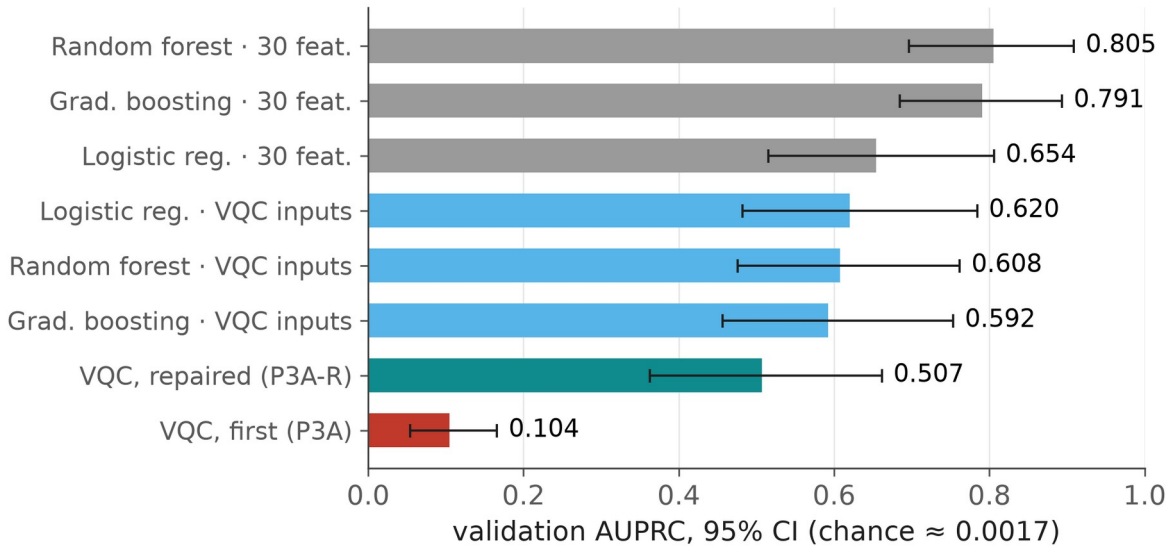
→

283,726 rows
473 frauds · 0.17% prevalence

→

Group-aware split
timestamp-only duplicates kept together

Partition	Rows	Frauds	Role
VQC train	127,677	213	fit every quantum model
VQC validation	28,372	47	qualification, thresholds
Controller train	42,559	71	fit mitigation policies
Controller validation	28,372	47	validate policies
Controller test	56,746	95	**sealed until the final test**

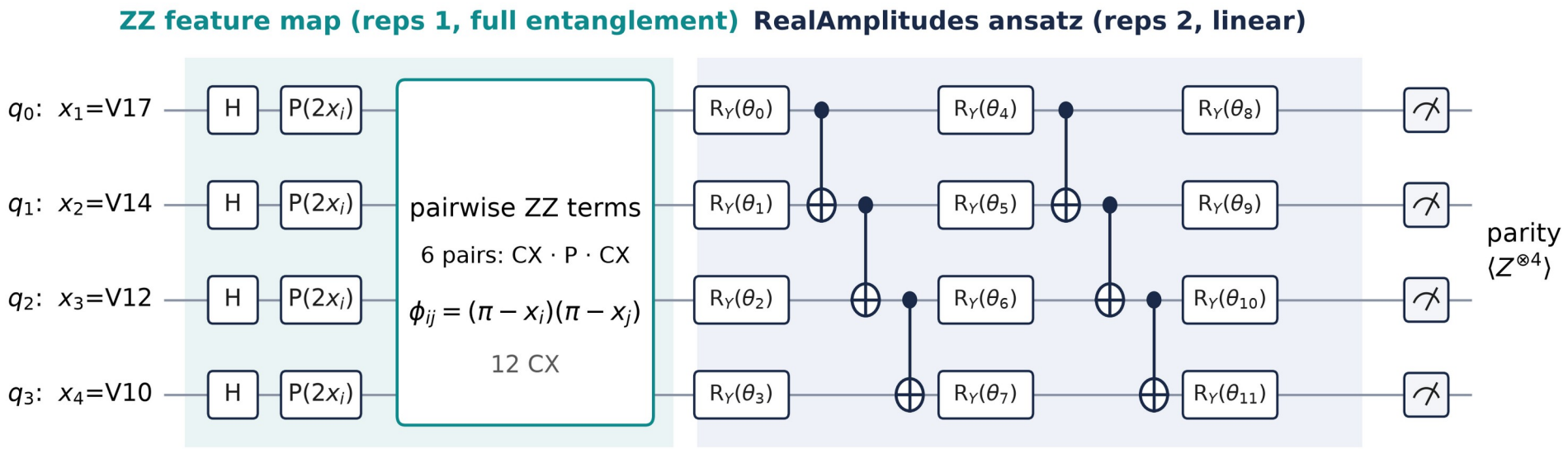


VQC inputs = the same 4 features and 1,065 training rows the quantum model uses.

Classical models set the bar: AUPRC 0.805 on all 30 features, 0.59–0.62 on the quantum model's four inputs.

Source: ulb_v1_dataset_summary.csv, ulb_v1_phase1_metadata.json, phase2_classical_metrics.csv, p3ar_validation_comparison.csv

The frozen quantum classifier



$p(\text{fraud}) = (1 - \langle Z^{\otimes 4} \rangle)/2$ · encoding e3: clip ±15 IQR → $x_i \in [-\pi/4, \pi/4]$ · 12 trainable parameters · 18 CX after transpilation

Inputs

Top-4 features by ANOVA F on training data (V17, V14, V12, V10), angle-encoded.

Training

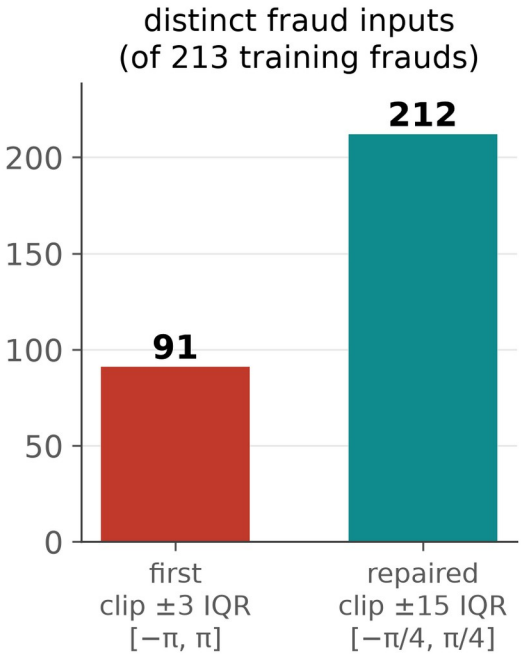
COBYLA, exact simulation, 1,065 rows: all 213 training frauds + 4 legitimate per fraud.

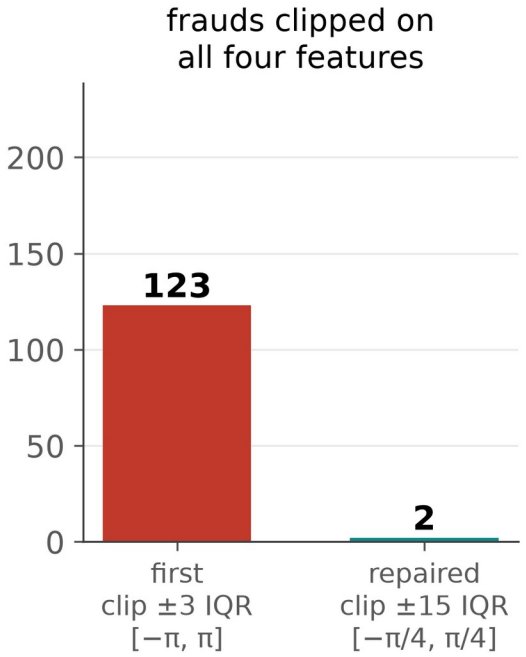
Frozen

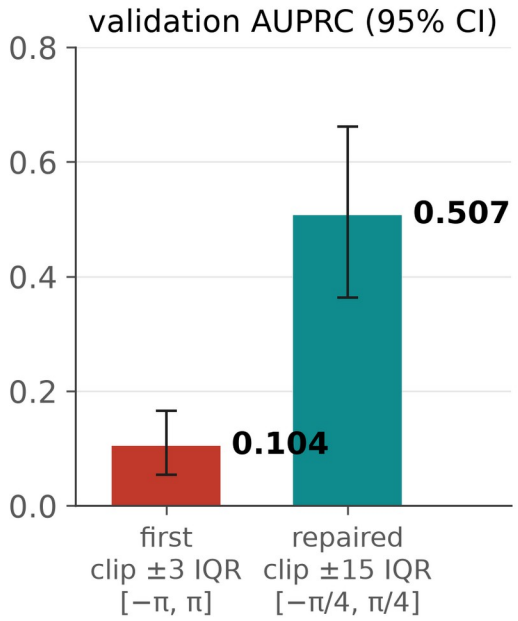
Parameters and circuit hashed; retraining is bitwise identical.

Source: configs/vqc/vqc3a_top4_ra1.yaml, configs/repair/p3ar_v1.yaml, p5a_pool_metrics.csv (anchor p3a_r)

Encoding repair: AUPRC 0.104 → 0.507







0.104 → 0.507

validation AUPRC (AUROC 0.957 after the repair)

91 → 212

distinct fraud inputs, of 213 training frauds

0.59–0.62

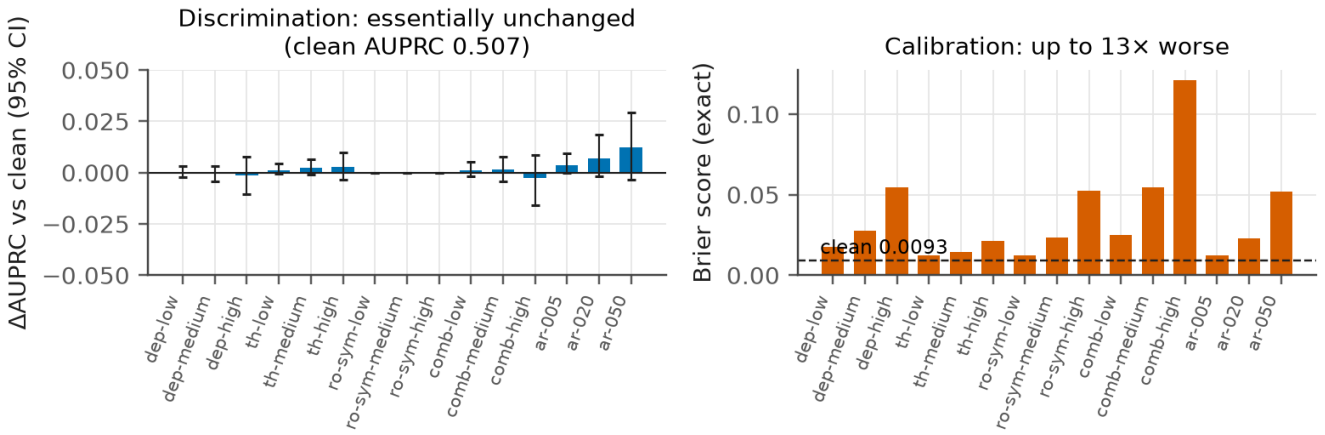
classical AUPRC on the same 4 inputs: no quantum advantage

A pre-registered, training-data-only repair fixed saturation and aliasing; the model works, but classical models stay ahead.

Source: p3ar_encoding_diagnostics.csv (encodings e0, e3), p3ar_validation_comparison.csv

First noise result, and where the project stands

Noise reshapes calibration, not discrimination (P3A-R on vqc_val; Phase 3B)



≤ 0.003
change in AUPRC

0.009 → 0.121
Brier, combined-high noise

0.53 → 0.03
precision at the frozen threshold

Project status

- **Oct 6** — Data, baselines, frozen VQC, repair
- **Oct 7** — Noise · mitigation · 64-model pool · categorical study
- **Oct 8** — Continuous-noise study · one-time sealed test
- **Oct 27** — Meeting: noise and mitigation in depth
- **Nov 10** — Meeting: policy selection
- **Nov 24** — Meeting: final findings, manuscript
- **Dec 8** — Final capstone defense

Noise mainly damaged calibration, not discrimination. Every experiment was complete by Oct 8; nothing is still running.

Source: p3b_exact_metrics.csv, p4_exact_diagnostics.csv; completion dates from the repository history